

Edge AI-Based UAV for Intelligent Fire Detection and Victim Prioritization during Emergency Response

DAI Project Report

Abstract

Fire emergencies in forests, industrial facilities, and urban environments often result in significant loss of life and property. Although Unmanned Aerial Vehicles (UAVs) have become increasingly popular for disaster monitoring, most existing systems are limited to fire detection and live video streaming, requiring rescue teams to manually interpret visual information and determine rescue priorities. This project proposes a low-cost Edge AI-based UAV capable of detecting fire, locating victims, estimating their observable condition, and prioritizing rescue operations in real time.

The proposed system integrates a Raspberry Pi 4 with a Raspberry Pi Camera Module V2, lightweight deep learning models, and a custom Context-Aware Victim Prioritization Engine (CVPE). The UAV continuously captures aerial video, detects fire and nearby humans, analyzes body posture and movement, estimates fire severity, and generates rescue priority scores based on observable contextual information such as victim pose, movement, and proximity to the fire. The generated mission report includes GPS coordinates and victim priority rankings, assisting emergency responders in making faster and more informed rescue decisions.

Unlike conventional UAV surveillance systems that merely identify fire locations, the proposed framework functions as an intelligent decision-support platform suitable for deployment on resource-constrained edge hardware. The system aims to improve situational awareness, reduce rescue response time, and provide an affordable solution for disaster management applications.

1. Introduction

Rapid disaster response is one of the most important factors influencing survival during fire emergencies. Fires occurring in residential buildings, industries, warehouses, forests, and public infrastructure create dangerous environments where firefighters often have limited visibility due to smoke, extreme temperatures, and structural instability. Entering these hazardous environments without prior situational awareness increases the risk to both victims and emergency responders.

Recent advances in Artificial Intelligence (AI), Computer Vision, and Edge Computing have enabled UAVs to perform more than simple aerial surveillance. Modern drones equipped with lightweight cameras and embedded computing platforms can process visual information in real time without requiring continuous cloud connectivity. This capability makes Edge AI particularly suitable for disaster management, where communication infrastructure may be unavailable or unreliable.

Most existing UAV-based emergency systems are designed primarily to detect fire or transmit live video streams to control stations. Although these systems successfully identify hazardous locations, they generally provide limited support for rescue planning. Emergency personnel must manually inspect video feeds to determine whether victims are present, assess their observable condition, and decide which rescue operations should be prioritized.

This project addresses this limitation by integrating fire detection, human detection, pose estimation, and intelligent rescue prioritization into a single Edge AI framework. The proposed system is implemented using a Raspberry Pi 4 and Raspberry Pi Camera Module V2 mounted on a quadcopter controlled by a Pixhawk flight controller. Instead of simply detecting fire, the system generates actionable information for rescue teams by identifying victims, estimating their observable condition, assigning rescue priorities, and producing GPS-tagged mission reports.

The proposed architecture is modular, allowing future integration of additional sensors such as thermal cameras, gas sensors, and autonomous navigation algorithms while maintaining compatibility with low-cost embedded hardware.

2. Problem Identification – Domain and Motivation

2.1 Application Domain

The proposed work belongs to the domain of Artificial Intelligence for Disaster Management, specifically focusing on Edge AI-enabled UAV systems for intelligent fire emergency response.

Fire incidents continue to pose serious risks to human life, infrastructure, and the environment. During such emergencies, rescue teams require rapid situational awareness to determine the severity of the incident, locate trapped victims, and allocate rescue resources efficiently. Conventional manual search operations are time-consuming and expose firefighters to significant danger.

Recent developments in UAV technology have enabled aerial monitoring of disaster sites from safe distances. Equipped with onboard cameras and embedded computing platforms, UAVs can quickly survey large areas and transmit visual information to emergency personnel. However, most existing UAV systems remain limited to detecting fire and providing live video streams.

2.2 Motivation

The primary limitation of current UAV fire monitoring systems is the absence of intelligent rescue decision support. Existing systems generally answer only one question: “Where is the fire?” They do not provide information such as:

- Are victims present?
- How many victims are detected?
- Which victims appear to require immediate rescue?
- Which victims are moving towards safety?
- Which rescue location should be prioritized?

As a result, firefighters must manually analyze drone footage under stressful and time-critical conditions. The proposed system addresses this gap by automatically combining multiple AI models to perform:

- Fire detection
- Fire severity estimation
- Human detection
- Pose estimation
- Victim tracking
- Rescue priority assessment
- GPS-based mission reporting

Rather than replacing emergency responders, the proposed system acts as an intelligent decision-support assistant that improves situational awareness and reduces cognitive workload during rescue operations.

2.3 Stakeholders

Primary stakeholders: Fire and Rescue Departments; National Disaster Response Force (NDRF); State Disaster Response Force (SDRF); Forest Departments; Industrial Emergency Response Teams; Municipal Fire Services.

Secondary stakeholders: local governments, disaster management authorities, smart city emergency control centres, and insurance agencies involved in disaster assessment.

2.4 Decision Supported

The proposed system supports operational decision-making during fire emergencies by assisting responders in answering: Has a fire occurred? What is the estimated severity of the fire? Are victims present? What is each victim's observable condition? Which victims should be rescued first? Where are the victims located? The system therefore provides decision support rather than simply reporting object detections.

3. Problem Statement

Current UAV-based fire monitoring systems primarily focus on detecting fire and transmitting aerial video, providing limited information regarding victim conditions and rescue planning. Given real-time RGB video streams captured using a Raspberry Pi Camera Module V2 mounted on a UAV, the proposed system aims to detect fire, estimate its severity, identify nearby victims, analyze their observable body posture and movement, and generate a Rescue Priority Score along with GPS-tagged mission reports. The system must operate under the computational constraints of a Raspberry Pi 4 while maintaining real-time inference performance despite challenges such as varying illumination, smoke, camera motion, aerial viewpoints, and small human objects. The proposed framework will be evaluated using Precision, Recall, F1-score, mAP@0.5, inference latency, frames per second (FPS), and Rescue Priority Classification Accuracy, and will be compared against conventional UAV systems that perform only fire detection without intelligent rescue prioritization.

4. Literature Survey

4.1 Introduction

The integration of Artificial Intelligence (AI) and Unmanned Aerial Vehicles (UAVs) has gained significant attention in disaster management due to their ability to rapidly monitor hazardous environments while minimizing risks to emergency responders. Recent advances in lightweight deep learning models and edge computing have enabled UAVs to perform real-time object detection and scene analysis directly on embedded platforms. Existing research has primarily focused on individual tasks such as fire detection, severity estimation, human/thermal detection, pose estimation, or aerial triage. However, relatively few studies combine these capabilities into a unified decision-support framework capable of assisting rescue teams during fire emergencies.

A total of 15 sources were reviewed for this survey. Eleven are drawn from the last three years (2023–2025) and published in reputed venues (IEEE, MDPI Q1/Q2 journals, Nature Scientific Reports, Frontiers). Three are conference papers or an unreviewed preprint, included strictly as supporting/contextual material. One older (2021) peer-reviewed journal source is retained as it is the only work in the pool that quantitatively addresses aerial consciousness/immobility classification relevant to victim-condition assessment.

4.2 Literature Comparison Table

Ref.	Year	Dataset	Method / Architecture	Key Metric & Value	Stated Limitation
[1]	2024	Custom aerial fire-image dataset (self-collected via quad-rotor UAV)	YOLOv8 / YOLOv5 detection + CNN-RCNN classification; onboard NVIDIA Jetson Nano	96% accuracy (YOLOv8 / CNN-RCNN classification); 89% (YOLOv8n / YOLOv5n detection)	Classification-only; no fire-severity or spread estimation; limited real-fire tests
[2]	2025	75,000+ image multi-source thermal SAR benchmark (10 sets + new mountain set, DJI M3T)	YOLOv8/9/10 & RT-DETR detectors; U-Net segmentation; benchmarked on Jetson AGX Orin	Comparative mAP / AR and energy-consumption benchmarking (single-class human detection)	Evaluated only on Jetson AGX Orin, not a Raspberry Pi-class board; daytime bias
[3]	2024	UAV-captured RGB + multispectral burnt-area imagery	Pixel-based SVM and Random Forest classifiers for burn-severity mapping	Random Forest: 95.5% overall accuracy (RGB burnt-area classification)	RGB alone performs poorly distinguishing severity classes; multispectral sensor is costlier
[4]	2025	Aggregated UAV/forest-fire benchmark imagery (review of lightweight CNN/YOLO variants)	Lightweight hybrids: MobileNetV3-YOLOv4, YOLOv8-SPDConv	MobileNetV3-YOLOv4: 94.6% acc., 37.2 ms/img; YOLOv8-SPDConv: mAP 89.5%	Gains reported on curated sets only; small/early-stage flame remains weakest case
[5]	2025	UAV wildfire imagery with occluded / small-scale flame instances	ResCaps-TC-Attn-Fire: Residual Capsule Network + Threshold Convolution + Attention	Improved robustness to occlusion/small flames vs. baseline CNN/YOLO (figures not itemized in abstract)	Inference cost not benchmarked on embedded/edge hardware
[6]	2024	15-subject pilot, 30 outdoor vital-sign measurements	UAS respiratory-movement detection + body-pose	89% overall triage-classification accuracy; F1-score = 0.94	Very small pilot sample; controlled daylight only; no smoke/occlusion or real casualties

Ref.	Year	Dataset	Method / Architecture	Key Metric & Value	Stated Limitation
		(simulated mass-casualty incident)	classification + drone heart-rate estimation		
[7]	2023	Simulated mass-casualty scene with volunteer casualties; 5G video link	OpenPose + YOLO posture recognition (7 posture classes) streamed over 5G	Reliable human detection and 7-class posture-based triage demonstrated (accuracy not itemized)	Healthy volunteers as casualties; posture-only triage, no vital-sign fusion; needs 5G link
[8]	2025	UAV aerial video benchmark for human activity recognition	YOLOv5 detection + pose estimation + LDA feature fusion	Improved activity/posture class-separability via LDA-enhanced pose features (accuracy not detailed)	Onboard real-time feasibility on low-power boards not evaluated
[9]	2025	Custom industrial tube-inspection image set	YOLOv8n / YOLOv8s deployed CPU-only on Raspberry Pi 500	YOLOv8n: precision 0.932, mAP@0.5 0.938, F1 0.914, ~470 ms/inference	Industrial-inspection domain, not fire/human; CPU-only inference too slow for multi-task pipeline
[10]	2025	Paired visible + infrared UAV forest-fire image set (novel fusion dataset)	Deep-learning visible-infrared image-fusion network for early fire detection	Proposes fused-sensor early-detection pipeline (benchmark not itemized in abstract)	Requires dual-sensor (RGB+IR) payload; visible-infrared registration remains challenging
[11]	2023	UAV-captured thermal night-time image set	Comparative YOLOv5 vs. YOLOv8 vs. SSD detectors on thermal imagery	YOLOv5 mAP@0.5 = 0.852; YOLOv8 mAP@0.5 = 0.855	Thermal-only, single context; no joint fire+human pipeline; not tested on low-power boards
[12]	2019	Custom UAV forest-fire aerial image set	YOLOv3 (small-scale CNN) on UAV onboard hardware [Conference paper]	83% recognition rate; 3.2 fps	Frame rate far below real-time needs; small network trades accuracy for speed
[13]	2020	Custom UAV post-disaster SAR thermal-image dataset	YOLOv3, YOLOv3-MobileNetV1/V3 (pruned & fine-tuned) [Conference paper]	Pruning/fine-tuning balanced speed vs. accuracy on embedded microcomputer (mAP not detailed)	Dataset limited to specific terrain types; small/distant humans reduce accuracy
[14]	2024	Simulated post-disaster search-and-rescue (PDSR) scenario/framework [arXiv preprint]	UAV deployment-optimization framework for hazard mapping and rubble-victim localization	Framework-level contribution; no field-benchmark accuracy figures reported	Unreviewed preprint; simulation-only, not validated on real imagery/hardware
[15]	2021	20-subject outdoor RGB + thermal dataset (simulated injured patients)	Mask R-CNN body detection + motion/feature-point tracking for consciousness/immobility classification	RGB: MCC 0.943, F1 0.951, AUC(PRC) 0.968; Thermal: MCC 0.913, F1 0.923, AUC(PRC) 0.960	Validated only with static ground cameras, not yet flown on a UAV

Note: metrics and limitations reflect what is explicitly reported in each source; where a numeric figure was not disclosed in the available abstract/text, this is stated rather than estimated.

4.3 Observations from Existing Literature

Fire Detection: Lightweight object detectors such as YOLOv5/YOLOv8 combined with CNN classification branches ([1]) and attention/capsule-based architectures ([5]) achieve high accuracy for fire presence detection, and are increasingly benchmarked directly on embedded boards such as the NVIDIA Jetson Nano ([1]). Severity assessment, however, is mostly handled as an offline burnt-area classification problem using RGB/multispectral imagery ([3]) rather than a real-time onboard signal.

Human & Thermal Detection: Thermal-image benchmarks ([2],[11]) and standard YOLO detectors show that human detection from aerial/UAV viewpoints is feasible with mAP@0.5 in the 0.85–0.86 range even on constrained hardware, though small and distant persons remain difficult and most large-scale benchmarking still targets Jetson-class rather than Raspberry Pi-class hardware.

Pose Estimation & Triage: Studies combining OpenPose/YOLO pose estimation with posture classification ([6],[7],[8],[15]) demonstrate that observable body posture and motion can support automated triage, achieving triage

accuracies around 89% and F1-scores above 0.9 in small pilot studies, but nearly all are validated only on healthy volunteers or static ground cameras under controlled daylight conditions.

Edge AI Deployment: Deployment studies on Raspberry Pi-class hardware ([9]) show that YOLOv8n can reach mAP@0.5 above 0.93 with sub-second CPU-only inference for a single detection task, but none of the reviewed papers benchmark a combined fire + human + pose + tracking pipeline on a Raspberry Pi 4 specifically.

Rescue Support: Although numerous studies perform fire detection, human detection, or triage individually, none of the 15 reviewed papers integrate fire severity, victim detection, posture, and rescue prioritization into a single onboard decision-support pipeline.

4.4 Research Gap

- Gap 1 – Independent AI tasks. Fire detection ([1]–[5],[10]–[12]), human/thermal detection ([2],[8],[9],[11],[13]), and triage/condition assessment ([6],[7],[15]) are addressed by separate, disconnected studies; none integrate all of these into a single UAV pipeline.
- Gap 2 – Limited edge-hardware validation. Only [9] reports concrete accuracy/latency on a low-power CPU-class board, and even that is for an unrelated industrial task; the fire and human/condition models elsewhere are validated on Jetson-class GPUs or static ground cameras, not a Raspberry Pi 4.
- Gap 3 – Small, controlled-condition pilots dominate triage research. [6] (15 subjects), [7] (healthy volunteers) and [15] (20 subjects, static cameras) all use small samples under clear daylight with no smoke or fire-scene occlusion, leaving robustness under real fire-emergency visual conditions largely untested.
- Gap 4 – Severity estimation is offline, not a live decision input. [3] and [4] treat fire/burn severity as a post-fire mapping or benchmark-accuracy problem rather than a real-time onboard metric feeding a rescue-priority score during an active fire.
- Gap 5 – No integrated prioritization or reporting. None of the 15 reviewed papers quantitatively couple detected fire severity with detected victim count/condition to produce an automated rescue-priority ranking or a structured, GPS-tagged mission report — the core contribution targeted by this project's CVPE.

4.5 Proposed Contribution

To address the identified research gaps, this project proposes a low-cost Edge AI framework for intelligent fire emergency response using a UAV. The system integrates lightweight computer vision models running on a Raspberry Pi 4 with a Raspberry Pi Camera Module V2 to perform real-time fire detection, human detection, pose estimation, and victim tracking. The primary novelty is the Context-Aware Victim Prioritization Engine (CVPE), which combines fire severity, victim pose, movement, and proximity to generate rescue priority scores. Unlike conventional UAV systems that simply detect hazards, the proposed framework acts as a decision-support tool by providing GPS-tagged mission reports and prioritized victim information to emergency responders. This integrated architecture is designed for deployment on affordable embedded hardware while maintaining real-time performance suitable for practical disaster management scenarios.

5. Proposed System Architecture

5.1 System Overview

The proposed system is a low-cost Edge AI-enabled Unmanned Aerial Vehicle (UAV) designed to provide intelligent decision support during fire emergencies. The system combines onboard image acquisition, lightweight deep learning models, contextual decision-making, and GPS-based mission reporting into a unified pipeline capable of operating on embedded hardware.

Unlike traditional UAV surveillance systems that simply identify fire locations or stream live video, the proposed framework performs multiple AI tasks sequentially to provide actionable information for emergency responders. The UAV continuously monitors the environment using an onboard camera, detects fire, estimates its severity, identifies nearby victims, analyzes their observable conditions, tracks their movement, and assigns rescue priorities before transmitting mission information to the ground station.

The complete architecture is designed to operate on a Raspberry Pi 4 connected to a Raspberry Pi Camera Module V2, while flight operations are controlled by a Pixhawk 2.4.8 flight controller. Communication between the Raspberry Pi and Pixhawk is achieved through the MAVLink protocol, enabling the AI system to access flight telemetry and GPS information during mission execution.

5.2 Overall System Pipeline

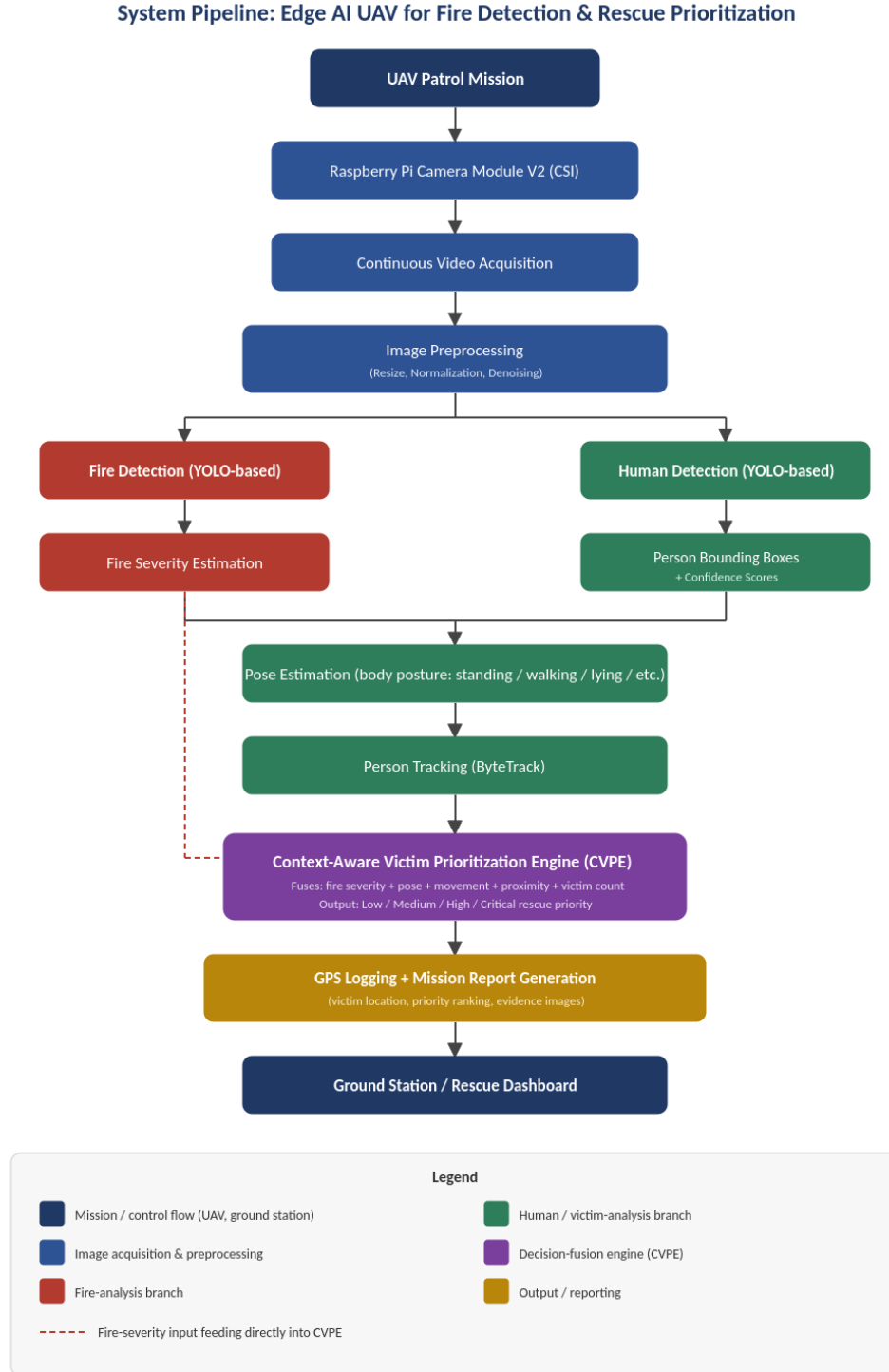


Figure 1. End-to-end onboard processing pipeline, from image acquisition through the Context-Aware Victim Prioritization Engine (CVPE) to the ground-station mission report.

The pipeline follows a strict acquire → detect → branch → fuse → report flow. After preprocessing, the frame is passed to two parallel branches — fire analysis (detection → severity estimation) and human analysis (detection → pose estimation → tracking). The fire-severity signal (dashed line in Figure 1) is fed directly into the CVPE alongside the tracked, pose-annotated victim data, so that rescue priority is always computed jointly from both branches rather than from either signal in isolation. The modular architecture ensures that each processing stage can be independently optimized or replaced without affecting the remaining components.

5.3 Hardware Architecture

The hardware architecture consists of five major components working together to enable autonomous surveillance and onboard AI inference.

Raspberry Pi Camera Module V2 – the primary vision sensor, continuously capturing high-resolution RGB video during flight. Compared to low-cost ESP32-CAM modules, it provides higher image quality, better dynamic range, and improved compatibility with real-time computer vision algorithms.

Raspberry Pi 4 – the companion computer responsible for all onboard AI computation: receiving frames, executing the deep learning models, processing detections, running the CVPE, and generating mission reports.

Pixhawk 2.4.8 Flight Controller – manages stabilization, waypoint navigation, altitude control, GPS positioning, and telemetry; communicates with the Raspberry Pi via MAVLink.

GPS Module – provides real-time geographic coordinates; whenever a victim is detected, latitude/longitude are recorded for rescue personnel.

Telemetry Module – a telemetry radio linking the UAV and ground station for mission status, GPS coordinates, and rescue information, without interfering with onboard AI processing.

5.4 Software Architecture

The software stack runs on Raspberry Pi OS using Python-based AI modules built on OpenCV and PyTorch, with flight communication via MAVLink and Mission Planner:

- Raspberry Pi OS, Python 3
- OpenCV, PyTorch
- YOLO11n (fire and human detection)
- YOLO-Pose / RTMPose (pose estimation)
- ByteTrack (multi-person tracking)
- MAVLink, Mission Planner (flight communication and mission control)

5.5 AI Processing Pipeline

Stage 1 – Image Acquisition: the Raspberry Pi Camera continuously captures RGB frames at a predefined frame rate.

Stage 2 – Image Preprocessing: resizing, pixel normalization, noise reduction, and colour-space conversion where required, to improve computational efficiency while preserving essential visual information.

Stage 3 – Fire Detection: a lightweight YOLO11n-based fire detector outputs a fire bounding box, confidence score, and location; if no fire is detected the UAV continues surveillance.

Stage 4 – Fire Severity Estimation: once fire is detected, severity is estimated from observable features such as flame size and affected image area, and classified as Low, Medium, High, or Critical for use by the CVPE.

Stage 5 – Human Detection: the fire region and surrounding area are analyzed by a YOLO11n human-detection model, producing bounding boxes and confidence scores for all visible people.

Stage 6 – Pose Estimation: each detected person is processed to determine observable body posture (standing, walking, running, sitting, lying), providing important contextual information about the victim's condition.

Stage 7 – Person Tracking: detected individuals are tracked across frames using ByteTrack to avoid duplicate counting, estimate movement, monitor behavioural change, and maintain victim identity.

5.6 Proposed Novelty – Context-Aware Victim Prioritization Engine (CVPE)

The Context-Aware Victim Prioritization Engine (CVPE) is the primary research contribution of this work. Instead of simply detecting victims, it combines multiple contextual observations — fire severity, victim pose, victim movement, distance from fire, number of nearby victims, and GPS location — to assign one of four rescue priority levels:

Low	Victim appears mobile and relatively safe.
Medium	Victim is close to the hazard but shows movement.
High	Victim shows limited movement and remains in a hazardous area.
Critical	Victim appears immobile or lying close to the fire and requires immediate attention.

Unlike previous approaches that stop at object detection, CVPE transforms multiple AI outputs into actionable rescue intelligence for emergency responders.

5.7 Output Interface

The processed information is presented through a mission dashboard displaying live UAV video, fire detection results and severity level, number of detected victims, rescue priority of each victim, GPS coordinates, mission timeline, and captured evidence images. At the end of each mission the system automatically generates a report containing the incident summary, victim details, GPS-tagged locations, and rescue priority rankings.

5.8 Design Justification

Raspberry Pi Camera V2	Higher image quality than ESP32-CAM; direct CSI interface for low-latency capture.
Raspberry Pi 4	Affordable edge computer capable of running lightweight AI models.
YOLO11n	Lightweight object detector with a good accuracy-speed trade-off for embedded deployment.
YOLO-Pose / RTMPose	Efficient estimation of human body posture from RGB images.
ByteTrack	Robust multi-object tracking with low computational overhead.
MAVLink	Standard communication protocol between companion computers and Pixhawk flight controllers.
CVPE	Integrates contextual information from multiple AI modules to support rescue decision-making rather than simple detection.

5.9 Expected Advantages

- Real-time onboard AI processing using edge computing.
- Simultaneous fire detection and victim localization.
- Context-aware rescue prioritization instead of simple fire alerts.
- Reduced dependence on cloud connectivity.
- Low-cost implementation using commercially available hardware.
- Modular architecture supporting future expansion with thermal cameras, gas sensors, or autonomous navigation.

6. Experimental Setup

6.1 Experimental Environment

The proposed system will be implemented on a custom-built quadcopter equipped with onboard edge computing hardware, to evaluate whether lightweight AI models can perform real-time fire detection, human detection, pose estimation, and rescue prioritization under practical operating conditions. All AI inference will be executed onboard the UAV using a Raspberry Pi 4 to minimize communication latency and improve reliability where internet connectivity may be unavailable.

UAV Frame	S500 Quadcopter Frame
Flight Controller	Pixhawk 2.4.8
Companion Computer	Raspberry Pi 4 Model B (4GB RAM)
Camera	Raspberry Pi Camera Module V2
GPS	Pixhawk GPS Module
Telemetry	915 MHz Telemetry Radio
Power Source	4S Li-Po Battery
Development System	Windows PC with Python

6.2 Software Environment

Raspberry Pi OS	Embedded operating system
Python 3	Development language
OpenCV	Image processing
PyTorch	Deep learning framework
YOLO11n	Fire and human detection
YOLO-Pose / RTMPose	Pose estimation
ByteTrack	Multi-person tracking
MAVLink	Flight communication
Mission Planner	UAV mission control

6.3 Datasets

Fire Detection	DFire
Fire Detection	FLAME
Human Detection	VisDrone
Human Detection	COCO
Pose Estimation	COCO Keypoints
Pose Estimation	MPII Human Pose
Action Recognition (optional)	Okutama-Action
Validation	Custom UAV videos

Datasets will be divided into a 70% training set, 15% validation set, and 15% testing set. The validation set will be used for hyperparameter tuning, while the testing set is reserved exclusively for final evaluation.

6.4 Evaluation Metrics

Fire Detection: Precision, Recall, F1-score, mAP@0.5.

Human Detection: Precision, Recall, mAP@0.5.

Pose Estimation: Percentage of Correct Keypoints (PCK), pose classification accuracy.

System Performance: Frames Per Second (FPS), average inference time (ms), CPU utilization, memory consumption.

Rescue Decision Evaluation (CVPE): Rescue Priority Classification Accuracy, average priority-assignment time, correct victim-ranking percentage.

6.5 Baseline Models

Baseline 1 – Fire Detection Only: Input → Fire Detector → Alert.

Baseline 2 – Fire + Human Detection: Input → Fire Detection → Human Detection → Alert.

Proposed System: Fire Detection → Fire Severity → Human Detection → Pose Estimation → Tracking → CVPE → Mission Report.

Performance improvements over these baselines will demonstrate the effectiveness of the proposed rescue-prioritization framework.

6.6 Ablation Studies

Experiment 1 – Fire Detection only: measures fire detection accuracy.

Experiment 2 – Fire Detection + Human Detection: evaluates victim localization.

Experiment 3 – Fire Detection + Human Detection + Pose Estimation: evaluates observable victim-condition estimation.

Experiment 4 – Complete Proposed Framework (Fire Detection → Human Detection → Pose → Tracking → CVPE): evaluates complete rescue-prioritization performance. Comparing these experiments demonstrates how each module contributes to overall system performance.

7. Feasibility Analysis

The proposed project is designed specifically to operate on affordable, commercially available hardware. The Raspberry Pi 4 provides sufficient computational capability to execute lightweight deep learning models such as YOLO11n and YOLO-Pose at usable frame rates for real-time demonstrations. Training will be performed on a desktop computer with GPU acceleration, while deployment will occur on the Raspberry Pi. Publicly available datasets eliminate the need for extensive data collection and reduce project development time. The modular architecture further improves feasibility because individual components can be developed and tested independently before complete system integration. The overall implementation is expected to be completed within eight weeks.

7.1 Risk Assessment

Low frame rate (multiple models simultaneously)	Use lightweight YOLO11n models; reduce image resolution; apply quantization if required.
Poor lighting (smoke, shadows, night)	Train on diverse datasets; apply image enhancement; test under varied lighting.
Small human objects at altitude	Use VisDrone for training; fly at optimized altitudes; fine-tune detection thresholds.
Camera motion / blur	Use stabilized flight; apply preprocessing; limit UAV speed during surveillance.
GPS inaccuracy (urban environments)	Average multiple GPS readings; record confidence values; allow manual verification via dashboard.

7.2 Fallback Plan

- Reduce input image resolution.
- Execute pose estimation only after a fire is detected.
- Skip action recognition and rely on pose estimation.
- Perform inference on a ground-station laptop while retaining onboard image acquisition.
- Replace computationally intensive models with smaller variants.

These alternatives ensure successful demonstration even under hardware limitations.

8. Expected Outcomes

- Detect fire in real time.
- Estimate fire severity.
- Detect multiple victims.
- Estimate observable body posture.
- Track victim movement.
- Assign rescue priorities.
- Record GPS-tagged victim locations.
- Generate mission reports for emergency responders.

Compared to existing UAV systems that provide only fire alerts, the proposed framework offers enhanced situational awareness and decision support.

9. Conclusion

This project proposes a low-cost Edge AI-enabled UAV capable of supporting emergency responders during fire incidents through intelligent perception and decision support. By integrating fire detection, fire severity estimation, human detection, pose estimation, victim tracking, and the proposed Context-Aware Victim Prioritization Engine (CVPE), the system extends the capabilities of conventional UAV surveillance from passive monitoring to active rescue assistance.

The use of a Raspberry Pi 4 and Raspberry Pi Camera Module V2 demonstrates that advanced AI applications can be deployed on affordable edge hardware without relying on continuous cloud connectivity. The modular architecture ensures scalability, allowing future integration of thermal imaging, autonomous navigation, gas sensing, and multi-UAV collaboration.

The expected outcome of this research is an efficient, practical, and economically viable decision-support system capable of improving situational awareness, reducing rescue response time, and assisting emergency teams in prioritizing victims during fire emergencies.

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